Implementation of Community Project for UG Program under 2025 Scheme (2025 batch and onwards)

Course Objective:

1. The Community Project / Societal Project is a practical, project-based learning course that develops students' awareness of Educational Social Responsibility (ESR) through meaningful engagement with community needs.
2. The project focuses on real-world societal issues, including adopting technology for rural and urban development, applying technological tools to improve governance in civic societies, and developing sustainable technological solutions to enhance quality of life.
3. Students may undertake the project in groups of up to four members. Including interdisciplinary topics, and are evaluated based on the prescribed assessment rubrics and learning outcomes**.**

Teaching Learning Process:
The suggested TLP for Community / Societal Project.
The following structured sequence of activities is to be carried out during the semester.

Week / Phase	Teaching–Learning Activity	Faculty Role	Student Activity	Output / Evidence
1	Course introduction, objectives, ethics, conduct, and assessment briefing	Explain course scope, rules, rubrics, and expected deliverables	Understand course requirements and form teams	Team list, course plan
2	Community problem identification and topic selection	Guide Topic Framing and Feasibility	Identify local issues through discussion and observation	Shortlisted project topic
3	Stakeholder interaction and preliminary survey	Demonstrate survey methods and stakeholder engagement	Collect initial data from community/users	Survey notes, interview record
4	Problem statement and objective formulation	Review and refine problem statement	Prepare problem statement, scope, and objectives	Approved problem statement
5	Literature/case study review and baseline analysis	Suggest relevant references and examples	Study existing solutions and compare approaches	Baseline analysis summary

6	Project planning and role assignment	Help teams organize tasks, timeline, and responsibilities	Plan activities, divide roles, and prepare schedule	Work plan / Gantt chart
7	Feasibility analysis and solution brainstorming	Check technical feasibility and practicality	Generate possible engineering solutions	Solution alternatives sheet
8	Selection of final solution and design approach	Review final concept and design logic	Finalize concept, methods, and expected outcome	Concept note / design outline
9	Data processing/calculations / sizing / mapping	Support Computations and technical review	Analyze data, perform calculations, prepare maps/charts	Technical analysis sheet
10	Prototype/model / audit / DPR / app development	Monitor progress and correct technical issues	Develop the selected output	Working model/draft DPR / map/app
11	Testing, refinement, and implementation planning	Evaluate technical correctness and improvements	Test prototype or validate proposal with users	Test results / revised output
12	Field implementation/demonstration / awareness activity	Facilitate community interaction and documentation	Conduct implementation or demonstration	Photos, attendance, implementation record
13	Documentation of outcomes, feedback, and impact	Review report structure and completeness	Compile findings, stakeholder feedback, and observations	Draft report
14	Presentation preparation and viva practice	Conduct mock presentation and feedback	Prepare slides/poster and rehearse	Presentation deck
15	Final presentation, submission, and reflection	Evaluate performance using rubrics	Present project, submit report, reflect on learning	Final report, presentation, viva

CIE Pass Rule:

- ✓ Passing Standard 50 marks out of 100.
- ✓ Use numerical scores from the rubric and keep brief evaluator comments for each criterion.
- ✓ For team projects: Record individual contribution within the documentation and adjust individual marks using a peer-evaluation sheet or guide's assessment.
- ✓ Maintain evidence (field notes, photos, questionnaires, Detail Project Reports) as annexures for CIE moderation.
- ✓ Use standardized score sheet (spreadsheet) mapping rubric items to columns for easy aggregation and archival

Methods for project identification:

Students may identify the project topic using one or more of the following methods (not limited to): Interaction with local rural or urban stakeholders such as local leaders, corporation/municipality/Panchayath officers, NGOs, etc.; review of issues highlighted in print and electronic media and social networks. After selecting a broad topic, collect data and requirements, conduct a feasibility survey, estimate timelines and resources, approximate expenditure (if applicable), and define the problem statement with objectives clearly.

List of Project Topics/Titles (Not Limited):

Sl. No.	Broad Areas/Project titles	Measurable engineering outcome	Specific POs	SDGs
1	Computer-Based Learning Tool for Mentally Challenged Students	Design and develop a web/browser-based learning tool as per the requirements of the special school.	5, 6	4
2	Medication management system for an old-age home.	Design and develop a web-based application to monitor the administration of medicine to senior citizens residing at an old-age home.	6	3
3	Virtual Labs for Learning Science	Design and Development simulation environment to do the experimentation list in Physics/Chemistry/Biology for high school in a rural area	5, 6	4
4	Local Skill Development and Employment Information Portal	Design and develop a web-based portal that provides information on local skill development programs, vocational training, internships, and employment opportunities for youth in the Community.	6	4, 8
5	Digital Complaint Portal for Gram Panchayat	Design and develop a web-based complaint portal (Specific nature) for Gram Panchayat	6	16
6	Plant Care Reminder Application	Implement a reminder mechanism that generates timely notifications for watering, fertilizing, pruning, and other plant-care activities.	6	13,15
7	Basic Crop Recommendation System	Develop a basic crop recommendation system that recommends suitable crops based on user-provided parameters such as soil type, season, rainfall, and water availability	6	2, 13, 15
8	Lost and Found Application	Develop a Lost and Found Application that enables users to report, search, and manage lost and found items	6	11, 16
9	Campus Navigation System	Develop a campus navigation system that enables users to locate important campus facilities and obtain directions between selected locations, which provides a simple digital map and location information for important facilities on campus.	6	15

RUBRICS FOR COMMUNITY PROJECT

TO BE EVALUATED BY THE INTERNAL COMMITTEE for 100 Marks

Parameters (Marks)	HIGH	MODERATE	LOW
Problem Definition & Need Assessment (Community / Field Survey) (10 Marks)	High (7–10) Community problem clearly defined through thorough field/literature survey; objectives and scope well articulated with strong relevance to society or environment.	Moderate (4–6) Problem definition is clear but survey evidence of relevance is limited; scope is only broadly stated.	Low (0–3) Problem definition is vague or irrelevant; little or no survey evidence provided; scope unclear.
Problem Statement Formulation, Approach & Feasibility (15 Marks)	High (11–15) Precise, innovative and feasible problem statement supported by a sound, well-structured engineering/community-service approach.	Moderate (6–10) Adequate problem statement; approach is feasible but has minor gaps.	Low (0–5) Statement is partial or marginally feasible; approach is flawed or largely missing.
Stakeholder Engagement, Data Collection & Analysis (15 Marks)	High (11–15) Thorough stakeholder engagement; comprehensive data collection with well-interpreted analysis.	Moderate (6–10) Adequate but limited engagement; data collected is reasonable but analysis is incomplete or superficial.	Low (0–5) Little or no stakeholder contact; no meaningful data or analysis presented.
Implementation & Execution of Solution / Intervention (20 Marks)	High (14–20) Solution or intervention fully implemented as planned, using appropriate tools, resources and domain knowledge; execution is efficient and well managed.	Moderate (8–13) Solution partially implemented; execution is functional but needs external support or has minor gaps.	Low (0–7) Implementation is incomplete or non-functional; execution shows major errors.
Impact, Outcome Analysis & Societal Benefit / Sustainability (15 Marks)	High (11–15) Clear, demonstrable societal/environmental benefit with a realistic, well-supported sustainability plan; outcomes validated.	Moderate (6–10) Benefit is plausible; sustainability considered with minor gaps; outcomes presented but not thoroughly validated.	Low (0–5) Benefit or sustainability is unclear or weak; no evidence of outcome validation.
Daily Work Log / Field Diary (10 Marks)	High (7–10) Log/diary is exceptionally detailed, consistently updated, well organized, with a complete record of field/practical activities.	Moderate (4–6) Log is adequately detailed but lacks consistency in entries; minor gaps in documentation.	Low (0–3) Log is missing significant content, poorly written, disorganized, or submitted late.
Project Report & Documentation (10 Marks)	High (7–10) Report formatted impeccably as per standards, with clear diagrams, proper references and well-structured sections.	Moderate (4–6) Report follows basic formatting guidelines with minor errors in structure, references or figures.	Low (0–3) Poorly structured report with major formatting errors, missing sections, or plagiarised content.
Presentation, Teamwork & Viva-Voce (5 Marks)	High (4–5) Confident, structured presentation with effective visuals; answers viva-voce queries fully; strong teamwork evident.	Moderate (2–3) Presentation is clear and satisfactory; responds to most queries; adequate teamwork.	Low (0–1) Presentation is poor or absent; unable to respond to queries; little teamwork evident.
Total CIE	**100**		

Assessment Structure

Continuous Internal Evaluation (CIE) – 100 marks

Students, working in batches of three to four, must undertake at least one project, or a combination of two related activities, aimed as far as possible at producing a logically implemented or implementable outcome. The CIE is conducted by a departmental committee consisting of the Head of Department or their nominee, the guide, and one senior faculty member, who assess the work using rubrics for a total marks of 100.

A student must secure a minimum of 50 marks out of 100 in the CIE as a passing standard. Students who fail to reach this minimum may be given additional opportunities to earn the required marks before the semester closes; however, no re-doing of the CIE is allowed once the semester has ended.

NOTE : There is no Semester End examination for the same as per the guidelines of DSCE

COE **Dean Academics** **Principal**